

generated by advanced computers that work at computing speeds thousands of times faster than PCs we use every day.

Three simple steps required to construct a holographic environment are:

1. The computer recognises the presence and movements of people and objects.
2. Then, it tracks those images.
3. Finally, it projects those images on a stereo-immersive surface.

The 3D reconstruction for tele-immersion is facilitated by projecting images on a stereo-immersive surface using IT. Two or more cameras take rapid sequential shots of the same object, continuously performing distance calculations and projecting those into the computer-simulated environment, to replicate real-time movement. By mixing cameras and the Internet telephony, videoconferencing allows real-time exchange of a vast amount of information effectively, without the need of the physical presence of every person in one central room.

The history

Ivan Sutherland, the great computer graphics pioneer, proposed the concept of Ultimate Display in 1965. It described a graphics display that allowed the user to experience a completely computer-rendered environment. In late 1960s, the ARPANET, conceived by the US Department of Defence, gradually evolved into the Internet. In 1998, Abilene, a backbone-research project was launched for the Internet2 research. The Internet2 needed an application that would challenge and stretch its networks capabilities. The head of Advanced Network and Services Inc. proposed tele-immersion as the application to drive forward the Internet2 research. Thus, the National Tele-immersion Initiative was formed.

The new broadband version of the Internet is capable of transmitting high volumes of data. Tele-immersion is the only application that can push



Fig. 2: A tele-immersive session in progress (Source: www.advanced.org/tele-immersion/news.html)

it to its boundaries. In May 2000, researchers at University of North Carolina (UNC), University of Pennsylvania and Advanced Networks and Services Inc. reached a milestone in developing this technology. A user sitting in an office at UNC in Chapel Hill, NC, was able to see life-like, 3D images of colleagues hundreds of kilometres away.

Elements of a holographic environment

Initial tele-immersive displays required users to put on unique goggles and a head device that tracked viewpoints of users looking at the screen. On the other end, people who appeared as 3D images were being tracked with an array of eight ordinary video cameras, while three other video cameras captured real light patterns projected in each room to calculate distances. This enabled proper depth to be recreated on the screen. Hence, if an observer moved his head to the left, he or she could see corresponding images that would be seen if he or she was actually in the room with the person on the screen.

Images on the screen were split and polarised to create a different image for each eye. The goggles then combined these images so that the brain recognised only one 3D image. This process was similar to how 3D movie glasses worked. Early experiments experienced some problems, often encountered during normal

videoconferencing. Scenes being projected were refreshed three times per second, which created an unsteady image. If that rate could be improved to ten frames per second, it would create a smooth projected image that would almost be like

looking at another person through a window. Scientists have been working on new technologies to support this type of communication. Here are some components and aspects of the new technologies:

Tele-cubicle. By means of which users communicate, a tele-cubicle consists of a stereo-immersive desk surface and two stereo-immersive wall surfaces (Fig. 2). These three display surfaces join to form a corner desk unit. Walls appear as windows to other users' environment, while desks join together to form a virtual conference table in the centre. This allows realistic inclusion of tele-immersion into the work environment as it consumes the average amount of desk space.

The Internet2. This will replace the current Internet infrastructure. It is a consortium made up of the USA government, industry and academia (over 180 universities) that was formed with the purpose of creating the Internet of the future. Good quality tele-immersion needs about 1.2Gbps. This new network will have a higher bandwidth and speeds 1000 times faster than today's Internet. This high bandwidth and speed network is necessary to transfer large amounts of data that tele-immersion will produce.

Display technologies. Stereo-immersive displays would have to present a clear view of the scenes being transmitted. Haptic sensors would allow people to touch pro-